De La Salle Holy Cross College

NOVEMBER EXAMINATIONS

2023

Grade 10

FURTHER STUDIES

MATHEMATICS

Examiner: A. Bux

Moderators: J. Blockley and J.P. Guilhermino

Time: 2 hours

Marks: 110

Name: Marking Guidelines final

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of **20** pages which include **11** questions. Please check that your question paper is complete.
2. Read the questions carefully.
3. **Answer ALL the questions on the question paper and hand it in at the end of the examination.**
4. Diagrams are not necessarily drawn to scale.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Ensure that your calculator is set in **DEGREE** mode.
7. Clearly show **ALL** calculations, diagrams, graphs, etc. that you have used in determining your answers. **Answers only will NOT necessarily be awarded full marks.**
8. Round off final answers to **THREE DECIMAL PLACES** unless otherwise stated.
9. It is in your own interest to write legibly and to present your work neatly; anything that is unclear will not be marked.
10. Two blank pages (9 and 20) are included in the paper. If you run out of space for a question, use these pages. Clearly indicate the question number of your answer should you use this extra space.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Q10	Q 11	Total
Mark												
Total	8	11	10	9	6	9	20	18	9	6	4	110

Remember that God is with you and will help and inspire you in all that you do

Formulae you may need:

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

SECTION A [44 Marks]

QUESTION 1 [8 Marks]

The following two-way contingency table summarises the results of all the contestants that phoned in for 'Montecasino What's in the Villa' competition on the 947 Drive Show with Thando.

	Female (F)	Male (M)	Total
Number of correct answers (C)	70	A	82
Number of incorrect answers (I)	B	32	42
Total	80	44	124

1.1 Write down the values of A and B. (2)

A = 12 ✓

B = 10 ✓

For the following questions, leave your answers in its simplest fractional form.

1.2 Calculate the probability that a randomly selected contestant:

(a) will be a Female who got the answer correct. (2)

$\frac{70}{124}$ ✓a = $\frac{35}{62}$ ✓ca

only simplified fraction form for Qu 1.2

(b) gets the answer wrong given that he is male. (2)

$\frac{32}{44}$ ✓a = $\frac{8}{11}$ ✓ca

penalise 1 if not in fractional form

(c) is a female given that she gets the answer correct? (2)

$\frac{70}{82}$ ✓a = $\frac{35}{41}$ ✓ca

QUESTION 2 **[11 Marks]**

Two events A and B are such that $P(A) = \frac{1}{2}$, $P(B) = \frac{3}{10}$ and $P(A|B) = \frac{1}{10}$.

2.1 Show that the probability that both events occur is $\frac{3}{100}$. (3)

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\frac{1}{10} = \frac{P(A \cap B)}{\frac{3}{10}}$$

✓ correct formula
✓✓ sub

$$\frac{3}{100} = P(A \cap B)$$

2.2 Calculate the probability that B occurs given that A has occurred. (3)

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{\frac{3}{100}}{\frac{1}{2}}$$ ✓ ✓

$$= \frac{3}{50}$$ ✓ ca or 0,06

2.3 Are A and B independent events? Support your answer using calculations. (3)

$P(A) \times P(B) = \frac{1}{2} \times \frac{3}{10}$ ✓a

$= \frac{3}{20}$

$P(A \cap B) = \frac{3}{100}$ ✓a

Since $P(A \cap B) \neq P(A) \times P(B)$ ✓

$\therefore$ A and B are not independent

or $P(A/B) = \frac{1}{10}$

$P(A) = \frac{1}{2}$

Since $P(A/B) \neq P(A)$

not independent

or $P(B/A) = \frac{3}{50}$

$P(B) = \frac{3}{10}$

Since $P(B/A) \neq P(B)$

$\therefore$ A & B not independent

2.4 Are A and B mutually exclusive events? Explain your answer. (2)

$P(A \cap B) = \frac{3}{100}$

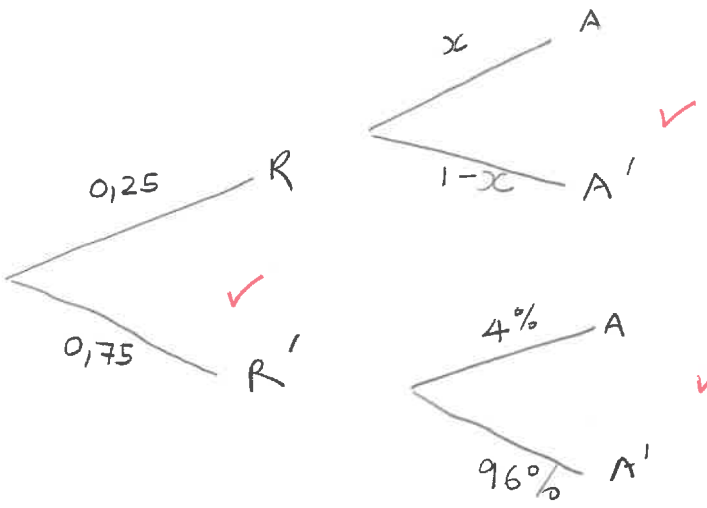
$\neq 0$ ✓

$\therefore$ not mutually exclusive. ✓

QUESTION 3 **[10 Marks]**

In the Eastern Cape, the probability of rain on any given day is 0,25. The probability that a car accident occurs on a sunny day is only 4%. The probability that an accident occurs on a rainy day is x. It is also known that the chances of a car accident on any given day is 1 in 5.

3.1 Illustrate the information in the form of a tree diagram and then calculate x. (7)



$P(A) = 0,25x + 0,75 \times 4\%$ ✓m ✓m

$0,2$ ✓a $= 0,25x + 0,03$

$0,68 = x$ ✓ca

3.2 Given that a car accident happened, determine the probability that it rained. (3)

$$P(R/A) = \frac{P(R \cap A)}{P(A)}$$

$$= \frac{0,25\ (0,68)}{0,2}$$ ✓ca ✓a

$0,25 \times 0,68 = 0,17$

$= 0,85$ ✓ca or $\frac{17}{20}$

QUESTION 4 **[9 Marks]**

Three sisters, Dakarai, Roxanne and Tamsin are to compete in a mathematics quiz with nine competitors in total.
Each of the nine competitors occupy one of nine cubicles numbered 1 to 9.

4.1 Write down the total possible number of arrangements of pupils in the nine cubicles at the start of the quiz. (2)

$9! = 362\ 880$ ✓ ✓a

4.2 Determine the total possible arrangements in which the sisters are all next to each other. (3)

_ _ _ _ _ _ _ _ _ _

$7! \times 3! = 30\ 240$ ✓ ✓ ✓ca

4.3 Determine the probability that Dakarai is in cubicle number 1, Roxanne in cubicle number 2 and Tamsin is in cubicle number 3. (4)

X X X _ _ _ _ _ _

$\frac{1 \times 1 \times 1 \times 6!}{9!}$ ✓ ✓ ✓

$= \frac{1}{504}$ ✓

or $\frac{1}{9} \times \frac{1}{8} \times \frac{1}{7}$ ✓ ✓ ✓

$= \frac{1}{504}$ ✓

QUESTION 5 **[6 Marks]**

For a specific set of data, [15;*a*] represents the interval of values within one standard deviation of the mean and [*b*;30] represents the interval of values within two standard deviations of the mean.

Calculate the values of *a* and *b*. (6)

$\bar{x} - \sigma = 15$ ✓m $-①$

$\bar{x} + 2\sigma = 30$ ✓m $-②$

① − ②

$-3\sigma = -15$

$\sigma = 5$ ✓ca

Sub $\sigma = 5$ in ①

$\bar{x} - 5 = 15$

$\bar{x} = 20$ ✓ca

$\therefore a = \bar{x} + \sigma$

$= 20 + 5$

$= 25$ ✓ca

$b = \bar{x} - 2\sigma$

$= 20 - 2(5)$

$= 10$ ✓ca

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL QUESTIONS ARE MARKED.

SECTION B **[66 Marks]**

QUESTION 6 **[9 Marks]**

Given $m = 4 + 3i$ and $n = 2 - 6i$.

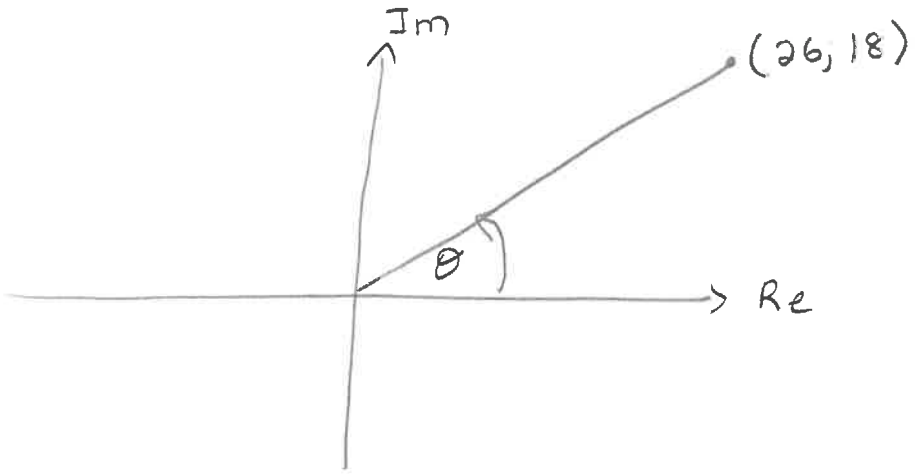
6.1 Determine the value of $m^*.n^*$. (3)

$(4-3i)(2+6i)$ ✓conj

$= 8 + 24i - 6i - 18i^2$ ✓prod

$= 26 + 18i$ ✓ca

6.2 Indicate $m^*.n^*$ on the Argand plane. (2)



✓quad (ca)
✓labels

6.3 Calculate the modulus and argument of $m^*.n^*$. (4)

$\arg(m^*.n^*) = \tan^{-1}\left(\frac{18}{26}\right)$ ✓m

$= 34,695^\circ$ ✓ca

modulus $= r = \sqrt{26^2 + 18^2}$ ✓m

$= 10\sqrt{10}$

$= 31,623$ ✓ca

QUESTION 7 **[20 Marks]**

7.1 Use the shortest possible method to simplify the following expression:

$(2x+3y+1)(2x-3y-1)$ (4)

$= [2x+(3y+1)][2x-(3y+1)]$ ✓ group

$= 4x^2-(3y+1)^2$ ✓ DOTS

$= 4x^2-(9y^2+6y+1)$ ✓ product

$= 4x^2-9y^2-6y-1$ ✓ ca

peralise 1 for not using short method

7.2 Determine the values of A, B, and C in the expression and its expansion.

$$(Ax+B)^5 = 32x^5 - 240x^4y^2 + 720x^3y^4 + C + 810xy^8 - 243y^{10}$$ (8)

$$(Ax+B)^5 = \binom{5}{0}(Ax)^5 + \binom{5}{1}(Ax)^4(B)^1 + \binom{5}{2}(Ax)^3(B)^2 + \binom{5}{3}(Ax)^2(B)^3 + \binom{5}{4}(Ax)^1(B)^4 + \binom{5}{5}(B)^5$$ ✓ma

$$= A^5x^5 + 5A^4x^4B + 10A^3x^3B^2 + 10A^2x^2B^3 + 5AxB^4 + B^5$$ ✓aa

$A^5 = 32$ ✓

$A^5 = 2^5$

$\therefore A = 2$ ✓

$5A^4x^4B = -240x^4y^2$ ✓

$5(2)^4x^4B = -240x^4y^2$

$B = \dfrac{-240x^4y^2}{5(2)^4x^4}$ $5\times2^4 = 80$

$= -3y^2$ ✓

$C = 10A^2x^2B^3$ ✓ or $\binom{5}{3}(Ax)^2(B)^3$

$= 10(2)^2x^2(-3y^2)^3$

$= 40x^2(-27y^6)$

$= -1080x^2y^6$ ✓

$(B)^5 = -243y^{10}$ ✓

$B^5 = (-3y^2)^5$

$B = -3y^2$ ✓

7.3 Simplify the following compound fraction:

$$\frac{\frac{\sqrt{9x^2-12x+4}}{2(3x-2)(2x+5)}+\frac{3}{3x-2}}{\frac{1}{3x-2}+\frac{1}{6(2x+5)}} \qquad (8)$$

$$= \frac{\frac{3x-2}{2(3x-2)(2x+5)} + \frac{3}{3x-2}}{\frac{6(2x+5)+3x-2}{6(2x+5)(3x-2)}}$$ ✓simp ✓ ✓ ✓LCD

$$= \frac{3x-2+6(2x+5)}{2(2x+5)(3x-2)} \times \frac{6(2x+5)(3x-2)}{6(2x+5)+3x-2}$$ ✓LCD ✓ ✓

$$= 3$$ ✓ca

$15x+28$

or

$$\frac{\frac{3x-2}{2(3x-2)(2x+5)} + \frac{3}{3x-2}}{\frac{6(2x+5)+3x-2}{6(2x+5)(3x-2)}}$$ ✓ ✓ ✓LCD

$$= \frac{3x-2+6(2x+5)}{6(2x+5)(3x-2)} \times \frac{6(2x+5)(3x-2)}{6(2x+5)+3x-2}$$ ✓

$$= 3$$

QUESTION 8 **[18 Marks]**

8.1 Factorise the following expression fully if $x \in \mathbb{C}$, without using a calculator. (4)

$x^2 + 8x + 25$

$= x^2 + 8x + 16 + 25 - 16$ ✓ma

$= (x+4)^2 - 9i^2$ ✓m

$= (x+4+3i)(x+4-3i)$ ✓✓

8.2 Solve for x, leaving your answers in **simplest surd form**. (2)

(a) $4x^2 - 4x - 1 = 0$

$x = \dfrac{-(-4) \pm \sqrt{(-4)^2 - 4(4)(-1)}}{2(4)}$ ✓ma

$= \dfrac{+1 \pm \sqrt{2}}{2}$ ✓ca

(b) $|x+7| = -2x - 1$ (6)

$x^2 + 14x + 49 = 4x^2 + 4x + 1$ ✓ma ✓ma

$3x^2 - 10x - 48 = 0$ ✓m

$(3x+8)(x-6) = 0$ ✓m

$x = -\frac{8}{3}$ ✓ca or $x \neq 6$ ✓a

Check: $x = -\frac{8}{3}$ LHS = RHS

$x = 6$ LHS ≠ RHS

or If $x+7 = -2x-1$ ✓ then $x+7 \geq 0$ ✓

$3x = -8$ $x \geq -7$

$x = -\frac{8}{3}$ Valid ✓a

If $-(x+7) = -2x-1$ ✓ then $x+7 < 0$ ✓

$-x-7 = -2x-1$ $x < -7$

$x = 6$ Not Valid. ✓a

8.3 Given $f(x) = x^3 - 36x - 2x^2 + 72$.

(a) Prove that $(x - 2)$ is a factor of $f(x)$. (1)

$f(2) = (2)^3 - 36(2) - 2(2)^2 + 72$

$= 0$ ✓

$\therefore (x-2)$ is a factor of $f(x)$

(b) Solve for x if $f(x) = 0$. (5)

$x^3 - 2x^2 - 36x + 72 = 0$

$(x-2)(x^2 + bx - 36) = 0$ ✓ ✓ ✓ fact any method

$(x-2)(x^2 - 36) = 0$ ✓ fact

$(x-2)(x+6)(x-6) = 0$

$x = 2$ or $x = -6; 6$ ✓ ca

$-2x^2 + bx^2 = -2x^2$

$bx^2 = 0x^2$

$\therefore b = 0$

or

$$x-2 \;\overline{)\; x^3 - 2x^2 - 36x + 72}$$

x^2 (quotient)

$x^3 - 2x^2$

$-36x + 72$

$-36x + 72$

or

$x^3 - 2x^2 - 36x + 72 = 0 \Rightarrow x^2(x-2) - 36(x-2)$ ✓ ✓

$x^3 - 36x - 2x^2 + 72 = 0$

$x(x^2 - 36) - 2(x^2 - 36) = 0$

$(x-2)(x^2 - 36) = 0$ ✓

QUESTION 9 **[9 Marks]**

9.1 The roots of a quadratic equation are given as:

$$x = \frac{-2 \pm \sqrt{1 - 3k}}{4}$$

(a) For which value(s) of k will the roots be real? (2)

$1 - 3k \geq 0$ ✓ma

$-3k \geq -1$

$k \leq \frac{1}{3}$ ✓ca

(b) Fow which value(s) of k will the roots be equal? (2)

$1 - 3k = 0$ ✓ma

$k = \frac{1}{3}$ ✓a

9.2 Given that $2x^2 + (p - 3)x + 8 = 0$, determine the value(s) of p for which the roots will be non-real. (5)

$b^2 - 4ac < 0$

$(p-3)^2 - 4(2)(8) < 0$ ✓ sub ✓

$p^2 - 6p + 9 - 64 < 0$

$p^2 - 6p - 55 < 0$ ✓ simp

$(p-11)(p+5) < 0$

CV : -5 ; 11



$-5 < p < 11$ ✓ ✓

QUESTION 10 **[6 Marks]**

Draw the graph of $y = -|2 - x| + 3$ on the axes provided.

- Determine the salient point and any intercepts.
- Clearly indicate the salient point and any intercepts on the axes. (6)

$y = -|2-x|+3$

$= -|x-2|+3$

Salient point : $(2;3)$

y-int : $y = -|0-2|+3$

$= -(2)+3$

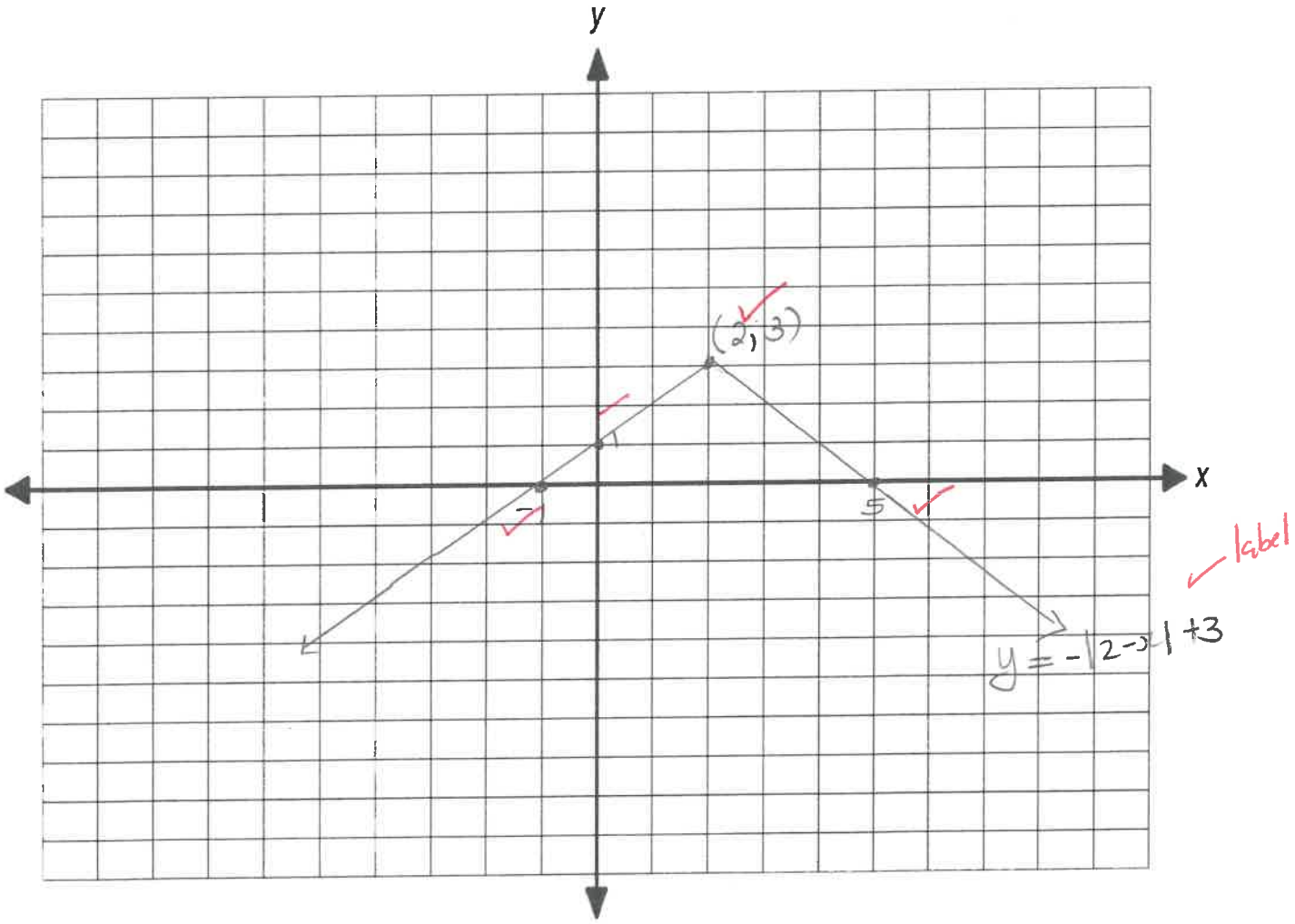
$= 1$

x-int : $0 = -|x-2|+3$

$+3 = +|x-2|$

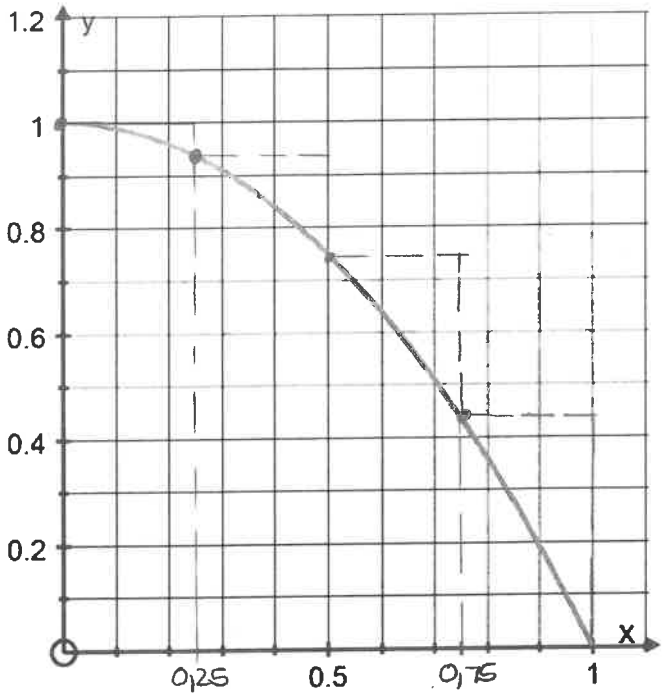
$\pm 3 = x-2$

$5; -1 = x$



QUESTION 11 **[4 Marks]**

Using left endpoint approximation, estimate the area under the curve of $f(x) = 1 - x^2$ for $0 \leq x \leq 1$. Use equally spaced partitions of 0,25.



(4)

Area $= \left[f(0) \times \frac{1}{4}\right] + \left[f(\frac{1}{4}) \times \frac{1}{4}\right] + \left[f(\frac{1}{2}) \times \frac{1}{4}\right] + \left[f(\frac{3}{4}) \times \frac{1}{4}\right]$

✓ correct width ✓✓ correct lengths using left end points

$= \frac{1}{4}\left[f(0) + f(\frac{1}{4}) + f(\frac{1}{2}) + f(\frac{3}{4})\right]$

$= \frac{1}{4}\left[1 + \frac{15}{16} + \frac{3}{4} + \frac{7}{16}\right]$

$= \frac{1}{4}\left[\frac{25}{8}\right]$

$= 0,781 \text{ units}^2$ ✓ ca

TOTAL: 110 marks

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL QUESTIONS ARE MARKED.